MDR Soil Moisture Modeling Report

Draft Research Summary

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March 23, 2026

Project:	Soil Moisture Modeling
Group:	MDR CS Group
Scope:	Base Model and Three-Regime Model
Status:	Internal Draft Report

Contents

1 Shared Methodology

1.1 Data

Data Sources

The modeling pipeline integrates satellite-derived variables, weather data, terrain information, and in-situ soil moisture observations. The target variable throughout the study is soil moisture measured at a depth of 5 cm.

Satellite Data

Satellite-based predictors were collected using Google Earth Engine [5]. The data sources include Sentinel-1, Sentinel-2, MODIS, SMAP, and a digital elevation model (DEM).

Sentinel-1 synthetic aperture radar (SAR) data [2] were used to extract VV and VH backscatter in both decibel and linear representations. Sentinel-2 Level-2A surface reflectance data [3] provided spectral bands $B2$, $B3$, $B4$, $B8$, $B11$, and $B12$. MODIS products [8] were used to obtain land surface temperature (LST) and NDVI. Terrain-related variables were derived from the SRTM DEM [12], including elevation, slope, and aspect. SMAP data [9] provided AM and PM soil moisture estimates along with retrieval quality flags. SoilGrids 2.0 [**?**] was used to collect the soil type.

To improve robustness and reduce noise, satellite values were aggregated over a spatial buffer around each station location. A buffer radius of 1000 m was used. The effective spatial resolution varies across datasets, with high-resolution products such as Sentinel-1 complementing coarser-scale inputs such as SMAP.

Weather Data

Weather variables were obtained from the Open-Meteo historical archive API [10]. This source was selected due to its availability of hourly observations without requiring extensive interpolation.

The primary variables used were rain and total precipitation (in millimeters). Hourly values were aggregated into daily totals before merging with station-level observations.

Ground Truth Soil Moisture Data

Ground-truth measurements were collected from in-situ monitoring stations. The temporal modeling experiments initially focused on five stations in Washington State.

For spatial generalization experiments, the dataset was expanded to include additional stations in climatically similar regions, including Idaho and Oregon.

For temporal modeling experiments, five stations in Washington State were used: Darrington-21-NNE and Quinault-4-NE (USCRN), and Spokane-17-SSW (USCRN), SourdoughGulch, and Touchet (SNOTEL). These stations span the state's major climate regimes, from the wet maritime conditions of the western Cascades and Olympic Peninsula to the semi-arid continental interior of eastern Washington.

The response variable for all supervised learning tasks is soil moisture measured at 5 cm depth.

External Test Datasets

Two types of external datasets were considered for evaluating generalization to new locations. The first consists of unseen public stations outside Washington with similar geographic and climatic characteristics. These datasets are used to evaluate temporal-spatial transfer. The second consists of ECE sensor deployments within Washington, which provide an additional real-world evaluation setting outside the original training stations.

TODO (ECE)

Temporal Coverage

Stage	Time Range
Initial target window	2008 – 2025
Aligned data window	2011 – 2025
Final study window (SMAP-constrained)	2017 – 2025

Table 1: Evolution of the temporal coverage as data sources were integrated.

The final study period (2017-2025) ensures consistent availability of all features, particularly SMAP-derived variables.

1.2 Preprocessing

Preprocessing Pipeline

The preprocessing pipeline transforms heterogeneous, multi-source data into a temporally aligned dataset suitable for supervised learning. It integrates satellite-derived features, weather variables, and in-situ measurements, while addressing differences in temporal resolution, missing data, and noise.

Pipeline Overview

Step	Description
Temporal alignment	Aggregate irregular observations to daily resolution
Spatial processing	Extract features using station-centered spatial buffers
Weather aggregation	Convert hourly weather data to daily totals
Imputation	Fill missing values using ensemble methods
Smoothing	Reduce noise in time series signals
Caching	Store intermediate results for efficiency

Table 2: Overview of the preprocessing pipeline.

Temporal Alignment

The final dataset is constructed at a daily resolution. However, satellite observations are often sparse and irregular in time. To address this, satellite data were retrieved in short temporal windows around each target date and aggregated into a single representative value.

For computational efficiency, data retrieval was performed at a weekly level, with aggregated values subsequently assigned to each day within the corresponding window. As a result, some features represent temporally aggregated estimates rather than exact daily measurements.

Spatial Processing

Satellite features were extracted using a spatial buffer around each station location, allowing each observation to represent an area rather than a single point. This improves robustness to noise and spatial misalignment between datasets.

A spatial buffer radius of 100 meters was applied for dynamic satellite products, aggregating pixel values within that footprint into a single representative mean.

Weather Aggregation

Weather data were obtained at an hourly resolution and aggregated into daily values prior to merging with station data. Rainfall and total precipitation were summed to produce daily totals (in millimeters), ensuring consistency with the temporal resolution of the dataset.

Imputation of Missing Values

Missing values arise due to differences in temporal resolution and data availability across sources. To address this, a temporal imputation system was applied after data merging and before feature engineering.

Each feature was processed independently. Time series were sorted, deduplicated, and validated before imputation. Missing values were reconstructed using an ensemble of interpolation-based, statistical, and machine learning methods.

Each imputer produces a prediction $\hat{x}_t^{(i)}$ and an associated confidence $c_t^{(i)}$. The final estimate is computed as a weighted combination:

$$\hat{x}_t = \frac{\sum_i w_t^{(i)} \hat{x}_t^{(i)}}{\sum_i w_t^{(i)}}, \quad w_t^{(i)} = b_i c_t^{(i)} \tag{1}$$

where b_i are predefined base weights. Outlier predictions are downweighted using a median absolute deviation (MAD) criterion [6], and confidence is adjusted based on the temporal distance to the nearest observed value.

The ensemble includes linear interpolation, forward/backward filling, rolling averages, seasonal climatology, regression models, Gaussian processes [11], and gradient boosted trees [1]. This combination enables robust reconstruction across both short gaps and long-term seasonal patterns.

Observed values are preserved, and only missing entries are filled. A binary mask is created for each feature to indicate whether a value was observed or imputed. Static terrain features are excluded from the ensemble and filled using forward and backward propagation.

Imputation uses only past and surrounding observations; future values are not used. Consequently, some missing values remain near dataset boundaries.

Smoothing and Feature Preparation

Following imputation, a smoothing step is applied to reduce noise in the time series. The resulting signals serve as inputs to the feature engineering stage.

Caching and Efficiency

Intermediate results from satellite and weather retrieval were cached to improve computational efficiency. Since the underlying data are static, caching avoids redundant data access without affecting feature values.

Temporal Coverage

The usable time window is determined by the intersection of all data sources. The inclusion of SMAP requires restricting the study period to January 1, 2017 through December 31, 2025, ensuring consistent feature availability.

1.3 Features

Feature Engineering and Selection

The feature set was constructed through a multi-stage process combining domain-driven feature engineering with data-driven selection and refinement. The objective was to capture the temporal dynamics of soil moisture while maintaining a compact and stable representation for modeling.

Feature Engineering Overview

Starting from the preprocessed time series, a large set of candidate features was generated by applying structured transformations to satellite, meteorological, and derived signals. These features were grouped into families based on the type of information they encode.

Family	Description
A (Change dynamics)	Differences, gradients, and percent changes capturing short-term wetting and drying
B (Volatility)	Rolling statistics and exponential moving averages capturing variability and trend
C (Memory)	Lagged observations and exponentially weighted memory terms
D (Seasonality)	Cyclical encodings and periodic descriptors
G (Meteorological)	Rainfall accumulation, API, and time-since-rain features
S (SMAP-derived)	Soil moisture signals and temporal transforms

Table 3: Feature families used in the engineering process.

This process resulted in a high-dimensional feature space (approximately 400 features) with intentional redundancy across temporal scales. While beneficial for representation learning, this redundancy requires controlled pruning during model construction.

Feature Pool Construction

The initial candidate set was constructed from the training split by removing the target variable and identifier columns. Features were then filtered using family-based inclusion rules, restricting the search space to engineered temporal features (families A, B, C, D, G, and S). This resulted in an initial pool of approximately 408 candidate features.

Model-Based Feature Pruning

Feature selection was performed using a two-stage wrapper method based on gradient boosted trees (XGBoost) [1]. At each pruning round, feature importance was estimated using permutation importance on the validation set, averaged across multiple random seeds to improve stability [4].

Let I_j denote the seed-averaged permutation importance of feature j. At round t, features were ranked by I_j, and the lowest-ranked fraction was removed:

$$F'_t = \mathrm{Top}_{n_{\mathrm{keep},t}}(I_j) \tag{2}$$

The candidate subset was accepted only if its validation R^2 remained within a predefined tolerance of the best score observed so far; otherwise, the update was rejected and the algorithm reverted to the previous best subset.

Stage	Description
Coarse pruning	Aggressive reduction to remove clearly uninformative features
Fine pruning	Conservative refinement with tighter performance constraints

Table 4: Two-stage feature pruning strategy.

This procedure reduced the feature set from approximately 408 candidates to a high-performing subset of roughly 100 features while maintaining or improving predictive performance.

Redundancy Reduction and Final Feature Set

Although the pruned feature set achieved strong validation performance, it still contained substantial redundancy, primarily due to high autocorrelation between features derived from similar signals at different temporal scales.

A final refinement step was therefore applied to remove highly correlated and near-duplicate features while preserving predictive signal. This step reduces overparameterization and improves model stability.

The final base model was trained using a reduced set of 38 features, retaining the most informative temporal and physical signals identified during pruning.

Final Feature Composition

Category	**Examples**
SMAP-derived	Smoothed moisture, AM/PM differences, gradients
Meteorological	Rainfall accumulation, API-based features
Temporal statistics	Rolling means, extrema, exponential averages
Lag features	Delayed system response signals
Seasonality	Sine/cosine encodings of annual cycle
Static features	Elevation, slope, aspect, soil fractions

Table 5: Composition of the final feature set (38 features).

This combination enables the model to capture both fast event-driven dynamics, such as rainfall-induced wetting, and slower processes, including seasonal trends and soil memory effects, while maintaining a manageable feature space.

2 Base Model

2.1 Training

Model Training

Model Choice

The base model was implemented using a gradient boosted decision tree framework (XGBoost) for regression. This choice was motivated by its ability to capture nonlinear relationships, handle heterogeneous feature types, and remain robust to feature scaling and moderate redundancy in the input space.

Alternative models, including Random Forest, LightGBM, and CatBoost, were evaluated during preliminary experiments but did not achieve comparable validation performance. As a result, XGBoost was selected as the final modeling framework.

The model predicts surface soil moisture at 5 cm depth, denoted as `soil_moisture_5cm`.

Data Splitting Strategy

Split	Time Range
Training	Jan 2017 – Dec 2020
Validation	Jan 2021 – Dec 2022
Test	Jan 2023 – Dec 2025

Table 6: Temporal data split used for model development and evaluation.

The dataset was partitioned using a strict time-based split to preserve temporal structure and avoid information leakage. No shuffling was applied, and all features were constructed using only past and contemporaneous observations.

Training Configuration

Parameter	Value
Objective	`reg:pseudohubererror`
Number of estimators	5500
Learning rate	0.04
Maximum depth	8
Subsample ratio	0.9
Column sampling	0.8
Minimum child weight	2
L2 regularization (λ)	1.5
L1 regularization (α)	0.03

Table 7: XGBoost training hyperparameters.

The pseudo-Huber objective was selected to improve robustness to outliers and heavy-tailed residuals, which are common in environmental time series.

Temporal Weighting

To account for temporal drift and emphasize more recent observations, a weighting scheme was introduced during training:

$$w_i = \exp\left(\beta \cdot (year_i - year_{\max})\right) \tag{3}$$

where $\beta = 0.2$ and $year_{\max}$ corresponds to the most recent year in the training data. The weights were normalized to have unit mean.

This formulation increases the influence of recent observations while preserving information from earlier periods. The weights were applied through the `sample_weight` parameter in XGBoost.

Unweighted vs Weighted Training

Model Variant	Description
Unweighted	Absolute error loss, no temporal weighting
Weighted	Pseudo-Huber loss with temporal weights

Table 8: Comparison of training configurations.

The weighted model consistently improved predictive performance on the held-out test set, achieving higher R^2 and lower error metrics. Consequently, the weighted configuration was selected as the final base model.

Training Objective and Evaluation

Model optimization was driven by minimizing the specified regression loss function. Model selection and comparison were based on validation and test metrics, including R^2, MAE, RMSE, bias, and quantile-based error measures.

The combination of a time-aware split, robust loss function, and temporal weighting enables the model to adapt to evolving environmental conditions while maintaining strong generalization performance.

2.2 Evaluation: Temporal Prediction

Evaluation

Evaluation Setup

Model performance was evaluated on a held-out test set covering the period January 2023 through December 2025. A strict temporal holdout was used, ensuring that all test observations occur after the training period.

Performance was assessed using multiple complementary metrics, including coefficient of determination (R^2), mean absolute error (MAE), root mean squared error (RMSE), unbiased RMSE (ubRMSE), bias, and quantile-based error statistics such as the 90th percentile absolute error.

Overall Performance

Metric	**Value**
R^2	0.822
MAE	0.0283
RMSE	0.0397
ubRMSE	0.0396
Bias	-0.0032
P90 $\lvert$error$\rvert$	0.0615

Table 9: Performance of the weighted base model on the test set (2023-2025).

These results indicate that the model captures a substantial portion of the variability in soil moisture while maintaining low absolute and squared error.

Effect of Temporal Weighting

To assess the impact of temporal weighting, the weighted model was compared to an unweighted baseline trained with identical hyperparameters.

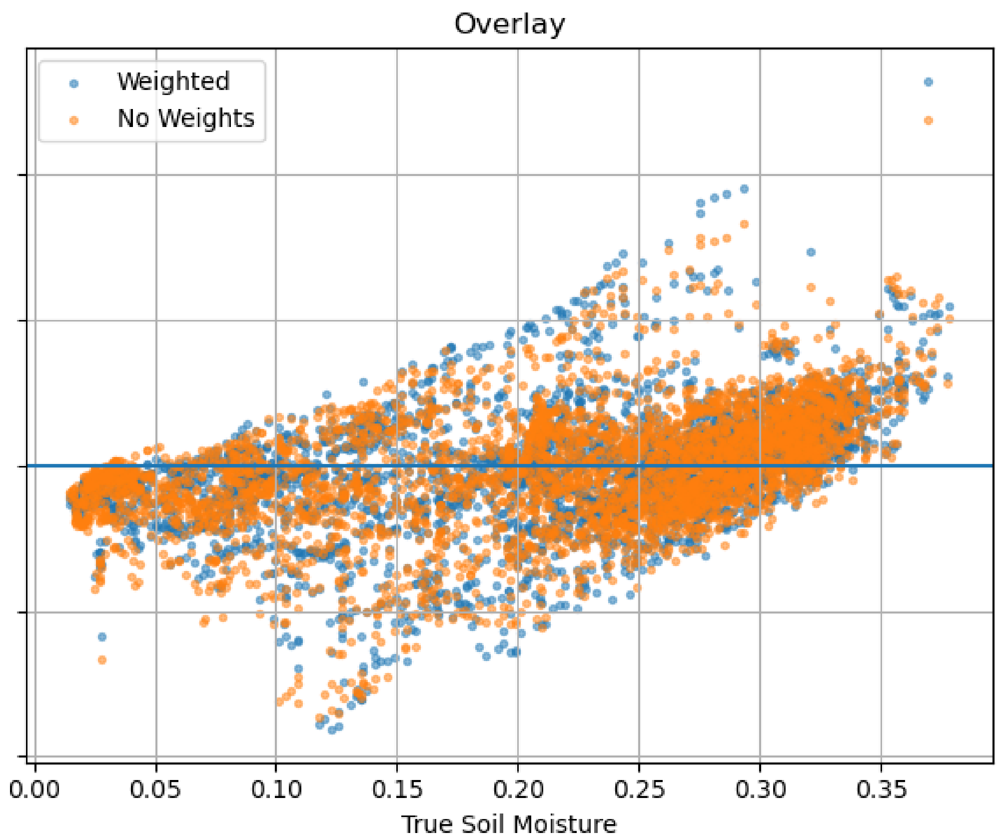

Figure 1: Comparison of predictions from weighted and unweighted models on a representative station. The weighted model shows slightly improved alignment with observed dynamics, although the overall residual structure remains largely unchanged.

Metric	Unweighted	Weighted
R^2	0.814	0.822
MAE	0.0304	0.0283
RMSE	0.0406	0.0397
ubRMSE	0.0405	0.0396
Bias	-0.0028	-0.0032
P90 |error|	0.0634	0.0615

Table 10: Comparison of unweighted and temporally weighted models.

Temporal weighting leads to consistent improvements across most metrics, including higher R^2 and lower error values. This suggests that emphasizing more recent observations improves generalization to future data, likely due to nonstationarity in environmental conditions.

Residual Analysis and Heteroskedasticity

Residual diagnostics revealed that model errors are not uniformly distributed across the target range. Both Breusch-Pagan and White tests indicate statistically significant heteroskedasticity (p-values $\ll 0.001$), confirming that error variance depends on soil moisture magnitude.

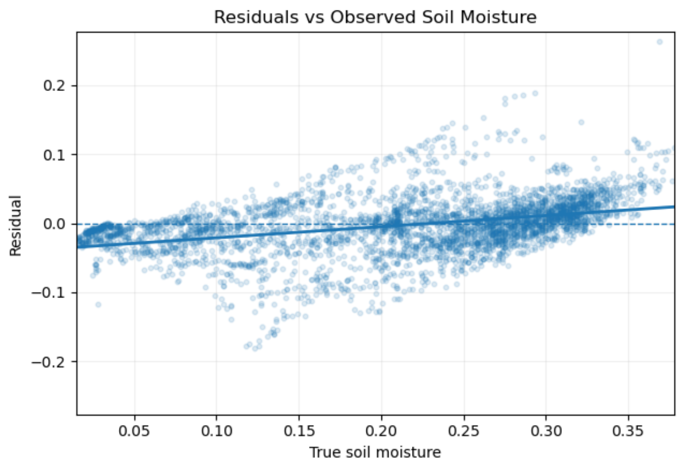

Figure 2: Residuals as a function of observed soil moisture. Error variance increases with moisture level, indicating heteroskedasticity.

Visual inspection further shows structured residual behavior, with increased variance in specific regions of the target space. In particular, prediction errors become larger and more dispersed at higher moisture levels, indicating that the model struggles more in mid-to-wet regimes.

This suggests that a single global model may not fully capture the variability of the underlying physical processes, particularly across different moisture conditions.

Regime-Based Performance

To further investigate error structure, the test data were partitioned into quantile-based bins according to the true soil moisture values, and model performance was evaluated within each bin.

The results show substantial variation across regimes. In particular, mid-range moisture conditions exhibit significantly worse fit compared to very dry conditions, with some bins producing strongly negative R^2 values.

This pattern indicates that the modeling task is inherently regime-dependent, with different physical processes dominating across moisture ranges. These findings directly motivate the exploration of regime-specific modeling approaches in the subsequent section.

2.3 Evaluation: Spatial Generalization

Evaluation Setup

In addition to temporal generalization, model performance was evaluated across spatial locations to assess the ability of the model to generalize to unseen stations. A leave-one-station-out (LOSO) cross-validation scheme was applied across the five Washington State stations used in this study: Darrington, Quinault, SourdoughGulch_WA_985, Spokane, and Touchet_WA_824.

For each fold, the model was trained on the combined training and validation data from the four remaining stations, and evaluated on all available rows from the held-out station. The same three-regime XGBoost ensemble architecture used in the temporal evaluation was applied without modification. No geographic stratification was applied; all five stations are located in Washington State with varying terrain, climate, and soil characteristics.

Results

Spatial evaluation results indicate how well the model transfers across locations with potentially different environmental characteristics, including soil composition, terrain, and climate.

Station	R^2	MAE	RMSE	ubRMSE	Bias
Darrington	0.730	0.0446	0.0543	0.0539	+0.0064
Quinault	0.498	0.0439	0.0529	0.0449	-0.0279
SourdoughGulch_WA_985	0.545	0.0518	0.0646	0.0591	+0.0262
Spokane	0.728	0.0499	0.0577	0.0570	-0.0092
Touchet_WA_824	0.403	0.0586	0.0677	0.0595	-0.0322
Mean	**0.581**	**0.0498**	**0.0594**	**0.0549**	–

Table 11: Per-station LOSO performance metrics for the base model.

Regime-level R^2 values are reported below for each held-out station.

Station	Dry R^2	Transition R^2	Wet R^2
Darrington	-0.400	-0.484	-9.992
Quinault	-0.547	-1.718	-1.833
SourdoughGulch_WA_985	-1.719	-2.217	-24.048
Spokane	-0.174	-1.263	-110.361
Touchet_WA_824	+0.143	-2.379	-154.901

Table 12: Per-station regime-level R^2 under LOSO evaluation.

Discussion

A key limitation of this evaluation is the small number of stations. With only five stations in total, the model is trained on four locations and tested on one — leaving it with very limited geographic diversity to learn from. As a result, the held-out station may have soil types, terrain, or climate conditions that are not well represented in the training data, making accurate generalization difficult regardless of model quality. Expanding the station network would be the most direct way to improve spatial performance.

Within these constraints, performance varies across stations. Darrington and Spokane transfer best ($R^2 \approx 0.73$), while Touchet_WA_824 shows the weakest performance ($R^2 = 0.40$). Biases differ in sign across stations, indicating that the model consistently over- or under-predicts at certain locations, likely because local soil or terrain conditions are not fully captured by the shared feature set.

Regime-level R^2 values are negative across all stations and moisture conditions. This reflects the same pattern noted in the temporal evaluation (Section 2.2): once data are restricted to a single moisture regime, the range of target values becomes narrow, and even small prediction errors are enough to produce a negative R^2. The wet regime is most affected because it has both the narrowest target range and the fewest samples. Overall, these results suggest that the model's spatial skill comes primarily from tracking broad moisture levels across locations, rather than capturing fine-grained variation within each regime.

2.4 Evaluation: New Locations

Evaluation Setup

To further assess generalization beyond the training and station-based evaluation settings, the model was evaluated on a set of entirely unseen sensor deployments (denoted as DEV3, DEV4, and DEV7). These devices represent independent measurement systems deployed at locations not included in any training, validation, or test splits.

Unlike the in-situ station data used throughout the primary evaluation, these sensors do not provide calibrated volumetric water content (VWC). Instead, they report soil moisture on a device-specific relative scale (min-max normalized), which is not directly comparable to the model's predictions in physical units.

To align with the daily temporal resolution of the modeling pipeline, hourly sensor readings were aggregated into daily values. This aggregation reduces noise but also results in a very limited number of usable observations per device.

As a result, absolute error metrics (e.g., RMSE, MAE, R^2) are not meaningful in this setting. To enable comparison, model predictions were normalized independently within each device to a relative scale. Evaluation was then performed using correlation-based metrics (Pearson and Spearman) and a directional consistency metric (trend agreement), with the goal of assessing whether the model captures coarse wet/dry temporal behavior rather than absolute magnitude.

Trend agreement is defined as the proportion of time steps for which the direction of change in the model prediction matches that of the observed signal:

$$\text{Trend} = \frac{1}{N-1}\sum_{t=2}^{N} \mathbb{1}\big(\text{sign}(\Delta\hat{y}_t) = \text{sign}(\Delta y_t)\big),$$

where $\Delta\hat{y}_t = \hat{y}_t - \hat{y}_{t-1}$ and $\Delta y_t = y_t - y_{t-1}$, and $\mathbb{1}(\cdot)$ denotes the indicator function.

Results

Summary metrics for each device are reported in Table 13.

Table 13: Relative evaluation metrics on unseen ECE devices

Device	n	Pearson	Spearman	Trend
DEV3	7	0.81	0.82	0.50
DEV4	8	0.65	0.55	0.57
DEV7	11	0.24	0.23	0.70

DEV3 exhibits relatively strong correlation (Pearson ≈ 0.81, Spearman ≈ 0.82), but this result is based on only seven observations and a highly unstable signal with a large dynamic range. DEV4 provides a more stable signal, but correlation drops to a moderate level (Pearson ≈ 0.65), and directional agreement remains weak. DEV7 shows very limited variability, leading to weak correlation (Pearson ≈ 0.24), while the higher trend agreement is likely driven by the flatness of the signal rather than meaningful alignment.

Discussion

The results indicate that the model exhibits some limited ability to track coarse wet and dry behavior at unseen locations. However, this conclusion is strongly constrained by the quality and quantity of the available data.

Across all devices, the number of usable observations is extremely small (7-11 daily samples per device), which severely limits the statistical reliability of all reported metrics. At this sample size, even relatively high correlation values can be driven by a small number of points and should not be interpreted as strong evidence. In addition, the lack of calibration in the sensor measurements prevents any evaluation of absolute prediction accuracy, restricting the analysis to relative comparisons only.

Device-specific characteristics further complicate interpretation. DEV3 shows high variability and produces seemingly strong correlation, but this is not defensible given the small sample size and unstable signal. DEV4 provides the most interpretable case, but the evidence remains limited to moderate correlation without consistent directional agreement. DEV7 exhibits minimal dynamic range, making it unsuitable for evaluating temporal behavior, as small fluctuations dominate the signal and can inflate directional agreement without reflecting meaningful alignment.

Overall, this evaluation does not provide strong or conclusive evidence of model performance on unseen locations. Instead, it serves as a limited qualitative check, suggesting that the model may capture some coarse temporal structure, while highlighting that the current ECE dataset is insufficient for rigorous validation. The primary evaluation of model accuracy therefore remains based on calibrated station data.

2.5 Observations

Temporal Characteristics

The following observations are based on the five Washington State training stations and reflect patterns identified during model development.

Soil moisture exhibits a clear seasonal pattern across stations, with higher values during winter months and lower values during late summer, as shown in Figure 3. While this general structure is consistent, the magnitude and shape of the seasonal cycle vary across locations, reflecting differences in local conditions and baseline moisture levels.

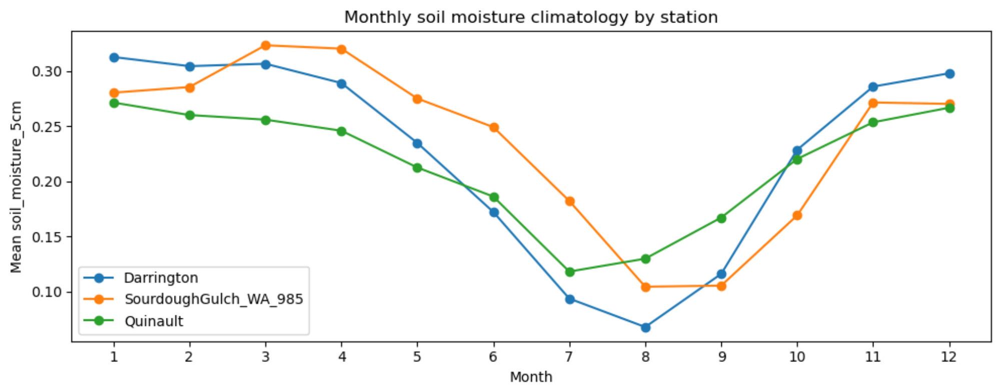

Figure 3: Monthly soil moisture climatology across selected stations.

At the daily scale, soil moisture exhibits strong temporal persistence, with relatively smooth evolution over time compared to the intermittency of precipitation. As shown in Figure 4, increases in soil moisture often follow rainfall events, while subsequent drying occurs more gradually over extended periods. This results in a characteristic pattern of relatively rapid increases followed by slower decay.

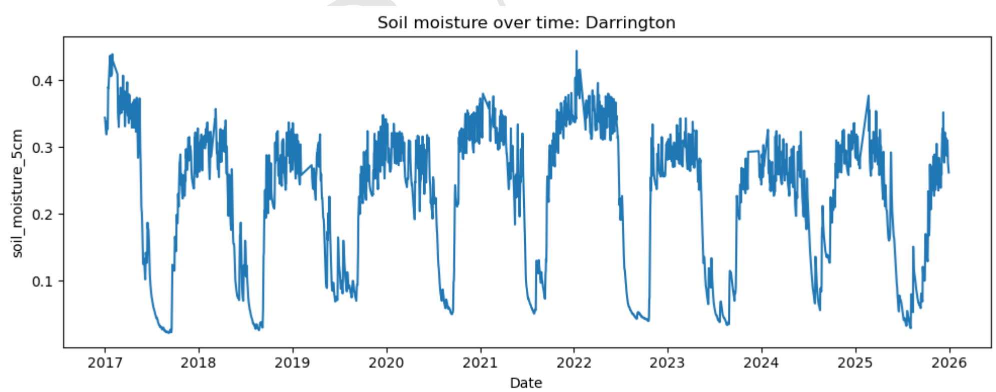

Figure 4: Daily soil moisture time series at the Darrington station.

These dynamics indicate that soil moisture is influenced by both short-term forcing (precipitation) and longer-term persistence, leading to lagged and smoothed responses. This behavior supports the use of temporal features such as lagged inputs and rolling statistics in the modeling pipeline.

Missingness Patterns

Feature availability varies across both time and data sources. Satellite-derived variables, particularly optical features, exhibit structured gaps likely driven by revisit frequency and environmental conditions such as cloud cover. These gaps occur in contiguous temporal segments rather than randomly, introducing discontinuities in the feature space.

In contrast, precipitation measurements and interpolated SMAP-derived soil moisture features are more consistently available over time. This results in uneven information density across features, requiring the model to rely more heavily on stable signals when optical data are unavailable.

Feature Behavior

Different feature groups exhibit distinct temporal characteristics. SMAP-derived soil moisture signals are relatively smooth and stable, capturing broad temporal trends with limited short-term variability. However, SMAP retrieval quality degrades in wet conditions, likely due to dense vegetation cover and standing water, which reduces its reliability precisely when moisture levels are highest. In contrast, optical features such as NDVI and Sentinel-2 bands show higher variability and occasional abrupt changes, as illustrated in Figure 5.

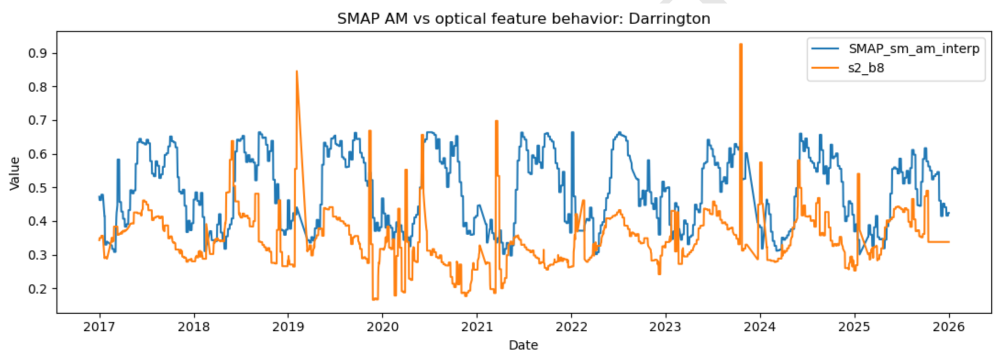

Figure 5: Comparison of SMAP-derived soil moisture and optical features.

Precipitation is highly intermittent, with sharp spikes corresponding to rainfall events. These events are associated with increases in soil moisture, followed by gradual declines over time. The resulting lagged relationship between precipitation and soil moisture is reflected in the engineered temporal features used in the model.

Dataset Challenges

Several characteristics of the dataset introduce challenges for modeling. First, soil moisture dynamics are nonstationary, with behavior varying across time and moisture conditions. This is reflected in the presence of distinct regimes (dry, transition, wet) identified in the model evaluation.

Second, residual analysis (Section 2.2) shows that prediction error is not uniform across moisture levels. Error variance is highest in the transition range (approximately 0.20–0.31 m^3/m^3), where competing processes such as wetting, drainage, and capillary rise occur simultaneously and are difficult to separate. This indicates that a single global model cannot perform equally well across all moisture conditions and motivates regime-aware modeling approaches.

In addition, the wet regime is underrepresented in the training data, comprising roughly 15% of observations. This, combined with the degraded satellite signal quality at high moisture levels, limits how much the model can learn about wet-regime dynamics. Variability across stations further introduces differences in baseline levels, dynamic range, and temporal patterns, making it difficult for a single global model to perform consistently across all conditions.

These observations motivate the use of temporal features, lag-based inputs, and regime-aware modeling strategies explored in subsequent sections.

3 Three-Regime Model

3.1 Training

Training Procedure

Modeling Objective

The three-regime framework was developed to evaluate whether regime-specific modeling can improve predictive performance relative to a single global model. This design was motivated by residual diagnostics from the base model, which indicated non-uniform error structure across the target space.

The target variable remains surface soil moisture at 5 cm depth, denoted as `soil_moisture_5cm`.

Regime Definition

Regimes were defined using thresholds derived from the empirical distribution of the target variable.

$$t_1 = 0.20, \qquad t_2 = 0.313 \tag{4}$$

Regime	Condition	Description
Dry	$y \le t_1$	Low-moisture, stable dynamics
Transition	$t_1 < y \le t_2$	Mixed persistence and variability
Wet	$y > t_2$	High-moisture, rapid dynamics

Table 14: Regime definitions based on empirical quantiles.

These thresholds serve as the primary regime boundaries throughout the analysis.

Model Architecture

The framework follows a hard-gated mixture-of-experts [7] architecture.

Component	Role
Anchor model	Trained on full dataset; provides global prediction
Transition specialist	Trained on transition-regime samples
Wet specialist	Trained on wet-regime samples
Dry regime	Handled directly by anchor model

Table 15: Components of the three-regime architecture.

A dedicated dry specialist was not retained in the final design. Empirically, the anchor model sufficiently captured dry-regime behavior, and introducing an additional model did not justify the added complexity.

Feature Usage

The anchor model is trained on a shared base feature set. Transition and wet specialists use augmented inputs that include both regime-specific features and the anchor prediction:

$$x_{\text{specialist}} = [x_{\text{regime}}, \hat{y}_{\text{anchor}}] \tag{5}$$

This formulation allows specialist models to learn corrections relative to a global estimate rather than reconstructing the full signal.

Training Configurations

Configuration	Description
Oracle-gated	Uses ground-truth regimes for routing; provides upper-bound performance
Self-gated (double-pass)	Uses anchor predictions for routing; deployable setting
Residual double-pass	Specialists predict residual corrections instead of absolute values

Table 16: Training configurations evaluated for the three-regime framework.

Oracle-Gated Training

In the oracle-gated setting, regime membership is determined using ground-truth values. The anchor model is trained on the full dataset, while transition and wet specialists are trained on their respective regime subsets.

During evaluation, samples are routed using true regime labels. Although not deployable, this configuration provides an estimate of the maximum achievable benefit under perfect gating.

Self-Gated Training

In the self-gated setting, regime labels are derived from model predictions rather than ground truth. To reduce leakage, pseudo-labels are generated using out-of-fold predictions from a time-series cross-validation scheme.

At inference time, the anchor model produces an initial prediction, which is used to assign regime membership. The corresponding specialist model is then applied.

Residual Specialist Training

In the residual variant, specialist models predict corrections to the anchor prediction:

$$r = y - \hat{y}_{\text{anchor}}, \qquad \hat{y} = \hat{y}_{\text{anchor}} + \hat{r} \tag{6}$$

This formulation simplifies the learning task by focusing each specialist on regime-specific residual patterns.

Temporal Structure

All experiments use a time-based split to preserve temporal ordering:

Split	**Time Range**
Training	Jan 2017 – Dec 2020
Validation	Jan 2021 – Dec 2022
Test	Jan 2023 – Dec 2025

Table 17: Temporal split used for three-regime experiments.

All training and gating procedures respect this temporal structure to prevent leakage.

Interpretation

The three-regime framework serves both as a predictive model and as an analytical tool. Its primary purpose is to test whether explicitly separating dry, transition, and wet dynamics can address the regime-dependent error structure observed in the base model.

While this decomposition aligns with the underlying physical intuition, its effectiveness ultimately depends on reliable regime identification and sufficient data within each regime.

3.2 Features

Feature Engineering and Regime-Specific Selection

The three-regime framework builds upon the feature engineering and selection pipeline used in the base model, while introducing regime-specific feature subsets tailored to different moisture conditions. The objective is to allow each specialist model to focus on the most relevant signals for its regime while retaining a shared global representation.

Shared Feature Engineering

As in the base model, candidate features were generated through structured transformations of satellite, meteorological, and derived signals. These include temporal differences, rolling statistics, lag features, seasonal encodings, and meteorological accumulations.

The initial feature space (approximately 400 features), along with the pruning and redundancy-reduction pipeline, is identical to the base model. This results in a compact set of informative features that serve as the foundation for the regime-based framework.

Anchor Feature Set

The anchor model operates on a shared base feature set consisting of 28 features selected from the pruned pool.

Category	Description
SMAP-derived	Soil moisture signals and temporal derivatives
Meteorological	Rainfall and API-based features
Temporal dynamics	Temperature and SAR-based transformations
Seasonality	Cyclical encodings and interaction terms
Static features	Terrain and soil properties

Table 18: Composition of the anchor feature set (28 features).

These features provide a general-purpose representation of soil moisture dynamics and are used to produce the initial global prediction.

Regime-Specific Feature Sets

Separate feature subsets were defined for different moisture regimes to enable specialization.

Regime	Focus	Feature Characteristics
Dry	Stable dynamics	Smoothed signals, temperature trends, static properties
Transition	Mixed behavior	Combination of persistence and short-term dynamics
Wet	Rapid dynamics	Variability, gradients, and higher-order temporal features

Table 19: Design of regime-specific feature sets.

The dry regime emphasizes stable and slowly evolving signals. In practice, a dedicated dry specialist model was not retained, as the anchor model sufficiently captured dry-regime behavior.

The wet regime incorporates higher-order temporal features, including rolling variance, range, percent change, and short-term gradients derived from SMAP, SAR, and meteorological variables. These features are designed to capture rapid changes and nonlinear responses.

The transition regime uses a similar feature space to the wet regime, reflecting the need to model intermediate dynamics where both persistence and rapid changes are present.

Anchor Conditioning

For transition and wet specialists, the anchor model prediction is included as an additional input feature. This allows specialist models to learn corrections relative to a global baseline rather than reconstructing the full signal.

$$x_{\text{specialist}} = \left[x_{\text{regime}}, \hat{y}_{\text{anchor}}\right] \tag{7}$$

This design provides contextual information about the global estimate and simplifies the learning task for regime-specific models.

Feature Usage Across Models

Model Component	Feature Set
Anchor model	Shared base feature set (28 features)
Transition specialist	Regime-specific features + anchor prediction
Wet specialist	Regime-specific features + anchor prediction
Dry regime	Handled by anchor model (no dedicated specialist)

Table 20: Feature usage across model components.

This structure introduces specialization only where necessary, while avoiding additional complexity in regimes where the global model already performs well.

Interpretation

The regime-specific feature design reflects the hypothesis that different physical processes dominate across moisture conditions. Dry regimes are characterized by slow dynamics and strong dependence on static properties, while wet regimes exhibit rapid and nonlinear behavior driven by short-term forcing.

Although this structured representation aligns with the underlying system dynamics, the evaluation results show that improved feature specialization alone is insufficient to guarantee better predictive performance under realistic gating conditions.

3.3 Evaluation: Temporal Prediction

Evaluation Setup

The three-regime framework was evaluated on the same temporal test set used for the base model (January 2023 through December 2025). Performance was compared across multiple configurations, including oracle-gated, self-gated, and residual-based variants, as well as the single-model anchor baseline.

Regime thresholds were defined using the 33rd and 66th percentiles of the target distribution:

Regime	Condition	Threshold
Dry	$y \leq t_1$	$t_1 = 0.20$
Transition	$t_1 < y \leq t_2$	$t_2 = 0.313$
Wet	$y > t_2$	–

Table 21: Regime definitions based on empirical quantiles of soil moisture.

Specialist Model Performance (No Gating) To separate model capacity from routing effects, each model was first evaluated independently on the full test set without regime-based gating.

Model	R^2
Dry specialist	0.33
Transition specialist	0.27
Wet specialist	-1.48
Anchor (base) model	**0.802**

Table 22: Global performance of individual models evaluated without gating.

The anchor model achieved the strongest overall performance. In particular, the dry specialist did not provide a meaningful improvement over the anchor despite being trained specifically on dry-regime samples.

Based on this observation, the anchor model was used in place of a dedicated dry specialist in all subsequent experiments. This simplifies the architecture while preserving performance in the dominant regime.

Oracle-Gated Analysis

An oracle-gated configuration was evaluated in which regime membership was determined using ground-truth values. This provides a diagnostic upper bound on performance under perfect routing.

Regime	**Samples**	**Model Used**	R^2
Dry	1509	Anchor	0.334
Transition	2015	Transition Specialist	0.148
Wet	492	Wet Specialist	-0.319

Table 23: Oracle-gated performance across moisture regimes.

Performance varies substantially across regimes. The dry regime, handled by the anchor model, shows moderate predictive power, while the transition regime is more difficult. The wet regime exhibits negative R^2 despite relatively small absolute errors, reflecting low variance and limited explainable signal.

To quantify the full potential of perfect routing, a global oracle-gated configuration was also evaluated:

Metric	Value
R^2	**0.859**
MAE	0.0255
RMSE	0.0354
ubRMSE	0.0342
Bias	-0.0090
Median AE	0.0184
P90 |error|	0.0578

Table 24: Global oracle-gated performance using ground-truth regime assignment.

This upper bound exceeds the anchor baseline ($R^2 \approx 0.802$) by a substantial margin, indicating that regime-specific modeling is beneficial when routing is accurate. The observed gap of approximately 0.06 in R^2 quantifies the cost of imperfect regime assignment.

Self-Gated Double-Pass

To approximate a deployable setting, regime assignment was based on model predictions rather than ground truth. Out-of-fold predictions from a time-series cross-validation scheme were used to construct pseudo-labels for specialist training, reducing leakage.

At inference time, samples were routed using anchor model predictions. This configuration achieved $R^2 = 0.766$, reflecting the difficulty of accurate gating under uncertainty.

Residual Double-Pass

A residual-based variant was also evaluated, in which specialist models predict corrections to the anchor model:

$$\hat{y} = \hat{y}_{\text{anchor}} + \hat{r}$$

This approach improved performance relative to the self-gated absolute model, achieving $R^2 = 0.801$, but did not surpass the anchor-only baseline. Regime-level analysis shows persistent instability in the wet regime, likely due to limited sample size and low variance.

Comparison to Anchor Baseline

Model	R^2
Self-gated (absolute)	0.766
Residual double-pass	0.801
Anchor-only baseline	**0.802**

Table 25: Comparison of three-regime variants with the anchor baseline.

Despite the additional modeling complexity, none of the mixture-of-experts variants outperform the anchor model. The anchor baseline remains competitive due to its ability to capture the dominant signal in the dry regime, where most observations are concentrated.

Discussion

The results confirm that soil moisture prediction exhibits strong regime-dependent behavior. However, they also demonstrate that hard-gated mixture-of-experts approaches do not automatically yield performance gains.

While oracle-gated results indicate clear potential, performance degrades under realistic gating. This highlights the importance of accurate regime assignment and suggests that future improvements should focus on more robust routing strategies, such as soft gating or probabilistic assignment.

Overall, while regime-aware modeling is well-motivated, effectively exploiting this structure in practice remains an open challenge.

3.4 Limitations

Limitations

Overview

The three-regime framework reveals meaningful structure in soil moisture dynamics, but several limitations affect its practical effectiveness. These limitations arise from data imbalance, model design choices, and evaluation constraints.

Key Limitations

Category	Description
Regime imbalance	The wet regime contains significantly fewer samples, leading to unstable model estimates and higher variance.
Low variance in wet regime	Small target variability causes R^2 to degrade even when absolute errors are small, complicating interpretation.
Gating error propagation	In self-gated settings, incorrect regime assignment leads to compounding errors through misrouted predictions.
Temporal pseudo-label constraints	Out-of-fold pseudo-labeling reduces effective training data, particularly in early time periods.
Model complexity vs performance	Increased complexity from specialist models and gating does not yield improved performance over the anchor baseline.
Hard gating assumption	Deterministic thresholds do not capture uncertainty near regime boundaries or mixed-behavior samples.
Threshold sensitivity	Fixed percentile-based thresholds introduce sensitivity to small distributional changes.
Limited spatial generalization	Overlap in station identities between splits limits conclusions about performance on unseen locations.

Table 26: Summary of key limitations in the three-regime framework.

Discussion

A primary challenge arises from imbalance and low variance in the wet regime. The combination of limited samples and narrow target range leads to unstable estimates and misleading evaluation metrics, such as negative R^2 despite low absolute error.

The effectiveness of the framework is also constrained by the gating mechanism. In realistic settings, regime assignment is imperfect, and errors in this stage propagate through the model, reducing overall performance. While oracle-gated results suggest potential gains, these are difficult to realize without reliable routing.

From a modeling perspective, the increased complexity of the mixture-of-experts framework does not translate into improved predictive performance. The anchor model already captures the dominant signal, particularly in the dry regime where most observations are concentrated.

Finally, the use of hard threshold-based gating and fixed regime boundaries introduces sensitivity and limits flexibility. These design choices do not fully capture transitional behavior or uncertainty near regime boundaries.

Implications

Overall, these results indicate that while soil moisture prediction exhibits clear regime-dependent structure, effectively leveraging this structure in a predictive model is nontrivial. Future work should focus on improved gating strategies, better handling of regime imbalance, and alternative formulations of regime-dependent learning.

References

[1] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD Conference*, 2016.

[2] European Space Agency. Copernicus sentinel-1 sar mission, 2024.

[3] European Space Agency. Copernicus sentinel-2 msi mission, 2024.

[4] Aaron Fisher, Cynthia Rudin, and Francesca Dominici. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81, 2019.

[5] Google Earth Engine Team. Google earth engine, 2024. https://earthengine.google.com/.

[6] Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 1974.

[7] Robert A. Jacobs and Michael I. Jordan. Adaptive mixtures of local experts. *Neural Computation*, 1991.

[8] NASA. Modis land products, 2024.

[9] NASA. Soil moisture active passive (smap) mission, 2024.

[10] Open-Meteo. Open-meteo weather api, 2024. https://open-meteo.com/.

[11] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[12] USGS. Shuttle radar topography mission (srtm), 2024.